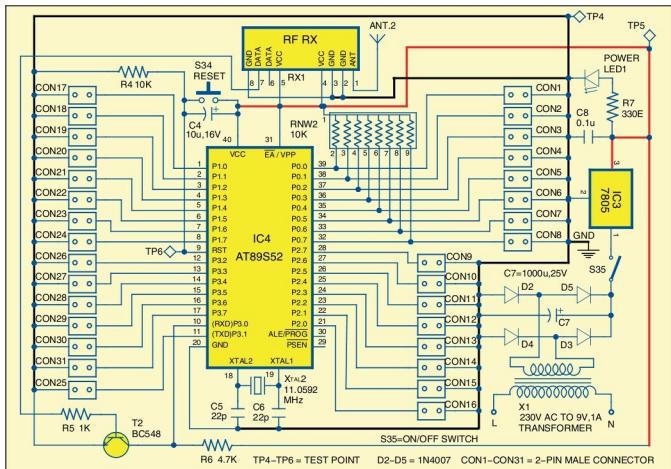




Ports P1, P2 and P3 are internally pulled-up bidirectional I/O ports and port P0 is an open-drain bidirectional I/O port. Port P0 is pulled high by resistor network RNW1. Port pins, P0.0 through P0.7, P2.0 through P2.7 and P1.0 through P1.7, are interfaced to switches S1 through S8, S16 through S9, and S17 through S24, respectively. Port pins, P3.0 and P3.2 through P3.7, are interfaced to S25 and S26 through S31, respectively.

Port pin P3.1 (TXD) is used for transmitting the signal. It is connected to transistor T1 biased in switched mode. For each button pressed, an ASCII code is transmitted via pin P3.1.

Power-'on' reset is provided by the combination of resistor R1 and capacitor C1. Switch S32 is used for manual reset. Crystal X_{TAL1} along with two 22pF capacitors provides basic clock frequency to microcontroller AT89552. IC 7805 (IC2) provides

regulated 5V supply to the circuit. Diode D1 acts as polarity guard. A 9V battery (BATT.1) is used to power the circuit.

Timer 1 is used for serial communication with 8-bit auto reload mode and 2400 baud rate at 11.0592MHz crystal. When any switch, say, S1 is pressed,

*Receiver and
appliance driver.*

Figs 2 and 3 show the circuits of RF receiver and appliance driver, respectively. This section comprises microcontroller AT89S52 (IC4), ASK receiver module (RX1), flip-flop CD4013 (IC5), three BC548 transistors (T2 through T4), relays (RL1

